

Hence, every cycle of length 3 is in S'_4

$$S'_4 = A_4$$

Possibilities: I, V_4, A_4

A_4 is non-abelian

$$A'_4 \neq \{I\}$$

$$|A_4/V_4| = 3$$

$\Rightarrow A_4/V_4$ is abelian

$$A'_4 \subseteq V_4$$

$\therefore A'_4 = V_4$ (as $A'_4 \neq \{I\}$)

Group Action

Let G be a group and A be a non-empty set. Then G is said to act on A if $\exists$ a func

$$*: G \times A \longrightarrow A$$

s.t.

$$1. g_1 * (g_2 * a) = (g_1 g_2) * a \quad \forall g_1, g_2 \in G, \forall a \in A$$

$$2. e * a = a \quad \forall a \in A$$

This map is known as a group action of G on A and A is called a G -set, we also say that G acts on A .

This action is also referred as action on A by left or left action of G on A .

We can also define action of G on A from right by considering mapping

$$*: A \times G \longrightarrow A$$

$$1) (a * g_1) * g_2 = a * (g_1 g_2)$$

$$2) a * e = a$$

Example:

1) Let $G, A (\neq \phi)$ s.t.

Define $*: G \times A \rightarrow A$ as

$$*(g, a) = g * a = a \quad \forall g \in G, \forall a \in A$$

Let $g_1, g_2 \in G$ & $a \in A$

then

$$g_1 * (g_2 * a) = g_1 * a = a$$

and

$$(g_1 g_2) * a = a$$

$$\Rightarrow g_1 * (g_2 * a) = (g_1 g_2) * a$$

and $e * a = a$

This shows that $*$ is a group action of G on A . This action is known as trivial action.

2) Take $A = G$

Define: $*: G \times A \rightarrow A$ as

$$g * a = ga \in A \quad \forall g \in G, a \in A = G$$

$$g_1 * (g_2 * a) = g_1 * (g_2 a) = g_1 g_2 a$$

$$(g_1 g_2) * a = g_1 g_2 a$$

$$g_1 * (g_2 * a) = (g_1 g_2) * a$$

$$e * a = ea = a$$

$*$ is a group action of G on $A(= G)$.

This action of G on itself is known as action by left or translation by left.

3) $*$: $G \times A \rightarrow A$

$$g * a = ag$$

$$g_1 * (g_2 * a) = g_1 * (ag_2) = ag_2g_1 \neq (g_1g_2) * a$$

So, $*$ is not a group action.

But

If we define $*$: $G \times A \rightarrow A$ as

$$g * a = ag^{-1}$$

then $\forall g_1, g_2 \in G, a \in A(= G)$

$$g_1 * (g_2 * a) = g_1 * (ag_2^{-1}) = ag_2^{-1}g_1^{-1} = a(g_1g_2)^{-1} = (g_1g_2) * a$$

$$e * a = ae^{-1} = a$$

$*$ is a group action and referred as action of G by right translation or right multiplication. This action is also known as regular action of G on itself.

4) Let G be a group. Take $A = G$

Define: $*$: $G \times A \rightarrow A$ as

$$g * a = gag^{-1} \quad \forall g \in G, \forall a \in A$$

Then $*$ is a group action on A and is known as action by conjugation.

Take $g_1, g_2 \in G$, and $a \in A(= G)$

$$\begin{aligned} g_1 * (g_2 * a) &= g_1 * (g_2ag_2^{-1}) \\ &= g_1g_2ag_2^{-1}g_1^{-1} \\ &= g_1g_2a(g_1g_2)^{-1} \\ g_1 * (g_2 * a) &= (g_1g_2) * a \end{aligned} \tag{1}$$

$$e * a = eae^{-1} \implies e * a = a \tag{2}$$

From (1) & (2), $*$ is a group action.

5) Let G be a group and $H \leq G$. Then take $A = G/H =$ set of all left cosets of H in G .

i.e. $A = \{aH \mid a \in G\}$

Define $*$: $G \times A \rightarrow A$ by

$$g * (xH) = gxH \quad \forall g \in G, xH \in A$$

Take $g_1, g_2 \in G, xH \in A$

Then

$$\begin{aligned} g_1 * (g_2 * xH) &= g_1 * (g_2xH) \\ &= g_1g_2xH \\ &= (g_1g_2) * xH \end{aligned}$$

$$e * xH = exH = xH$$

$*$ is a group action.

6) Let G be a group and A be set of all subgroups of G . Define

$$* : G \times A \rightarrow A$$

as

$$g * H = gHg^{-1} \quad \forall g \in G, \forall H \in A$$

Then $*$ is also a group action.

$$\forall g_1, g_2 \in G, H \in A(\leq G)$$

$$\begin{aligned} g_1 * (g_2 * H) &= g_1 * (g_2 H g_2^{-1}) \\ &= g_1 g_2 H g_2^{-1} g_1^{-1} \\ &= (g_1 g_2) H (g_1 g_2)^{-1} \\ g_1 * (g_2 * H) &= (g_1 g_2) * H \\ e * H &= e H e^{-1} = H \end{aligned} \tag{1}$$

From (1) & (2)

• **Remark:** If $* : G \times A \rightarrow A$ is an action then

if $x, y \in A$ such that $\underline{x = y}$

then $\forall g \in G$

$$(g, x) = (g, y)$$

$\Rightarrow g * x = g * y$ (as $*$ is a map)

$$\forall x, y \in A \text{ s.t. } \underline{x = y} \Rightarrow g * x = g * y \quad \forall g \in G$$

Theorem: Let G be any group and A be a non-empty set. Then any homomorphism from $G \rightarrow \text{Sym}(A)$, the symmetric group of A , defines an action of G on A .

Conversely, every action of G on A induces a homomorphism from $G \rightarrow \text{Sym}(A)$.

Proof:

Let $f : G \rightarrow \text{Sym}(A)$ be a group homomorphism.

Let for $g \in G$, $f(g) = \sigma_g$, i.e. $\sigma_g : A \rightarrow A$ is a bijective map.

For $g_1, g_2 \in G$

$$\begin{aligned} f(g_1 g_2) &= f(g_1) \circ f(g_2) \\ \sigma_{g_1 g_2} &= \sigma_{g_1} \circ \sigma_{g_2} \end{aligned}$$

Defined $* : G \times A \rightarrow A$ as

$$g * a = \sigma_g(a), \quad \forall a \in A$$

then $*$ is a map.

For $g_1, g_2 \in G, a \in A$

$$\begin{aligned} g_1 * (g_2 * a) &= g_1 * (\sigma_{g_2}(a)) \\ &= \sigma_{g_1}(\sigma_{g_2}(a)) \\ &= (\sigma_{g_1} \circ \sigma_{g_2})(a) \\ &= \sigma_{g_1 g_2}(a) \\ g_1 * (g_2 * a) &= (g_1 g_2) * a \end{aligned}$$

$$f(e) = \sigma_e$$

So, $f(e) = I_A$ as f is group homomorphism

$$e * a = \sigma_e(a) = I_A(a) \implies e * a = a$$

Therefore, $*$ is a group action of G on A .

Conversely, define $f : G \rightarrow \text{Sym}(A)$ as
for $g \in G$, $f(g) = \sigma_g \in \text{Sym}(A)$
 $\sigma_g : A \rightarrow A$ is defined as

$$\sigma_g(a) = g * a \quad \forall a \in A$$

First we show that f is a map and for this we have to prove that σ_g is a bijective map.
Clearly, σ_g is a map.
Let $a_1, a_2 \in A$ s.t.

$$\begin{aligned} \sigma_g(a_1) = \sigma_g(a_2) &\Rightarrow g * a_1 = g * a_2 \\ &\Rightarrow g^{-1} * (g * a_1) = g^{-1} * (g * a_2) \\ &\Rightarrow (g^{-1}g) * a_1 = (g^{-1}g) * a_2 \quad (\text{as } * \text{ is an action}) \\ &\Rightarrow e * a_1 = e * a_2 \quad (\text{as } * \text{ is a group action}) \\ &\Rightarrow a_1 = a_2 \end{aligned}$$

$\therefore \sigma_g$ is a one-one map.
Let $a \in A \Rightarrow g^{-1} * a \in A$ and

$$\begin{aligned} \sigma_g(g^{-1} * a) &= g * (g^{-1} * a) \\ &= (gg^{-1}) * a \\ &= e * a = a \end{aligned}$$

σ_g is onto map, and hence bijective.
Thus σ_g is in $\text{Sym}(A)$, hence f is a map.
Let $g_1, g_2 \in G$ then

$$\begin{aligned} f(g_1g_2) &= \sigma_{g_1g_2} \\ &= \sigma_{g_1} \circ \sigma_{g_2} \\ f(g_1g_2) &= f(g_1) \circ f(g_2) \end{aligned}$$

$$\sigma_{g_1g_2} : A \rightarrow A$$

$$\begin{aligned} \sigma_{g_1g_2}(a) &= (g_1g_2) * a \\ &= g_1 * (g_2 * a) \\ &= g_1 * \sigma_{g_2}(a) \\ &= \sigma_{g_1}(\sigma_{g_2}(a)) \\ \sigma_{g_1g_2}(a) &= (\sigma_{g_1} \circ \sigma_{g_2})(a) \end{aligned}$$

$\sigma_{g_1g_2} = \sigma_{g_1} \circ \sigma_{g_2}$
Hence, f is a group homomorphism.

Remark: For any group action of a group G on a set A , there is a homomorphism

$$f : G \longrightarrow \text{Sym}(A)$$

This homomorphism sometimes called the associated permutation representation of the given action.

Corollary: (Cayley's Theorem) Every group is isomorphic to a subgroup of its permutation group or symmetric group.

Proof: We know that any group act on itself under the action $*$ defined as

$$g * a = ga \quad \forall g \in G, a \in A \quad (\text{here } A = G)$$

$$*: G \times A \rightarrow A$$

then by previous theorem, the corresponding permutation representation to this action is given by homomorphism